FIRST EVER Eu3+ COMPLEX AS MEMBRANE PROBE

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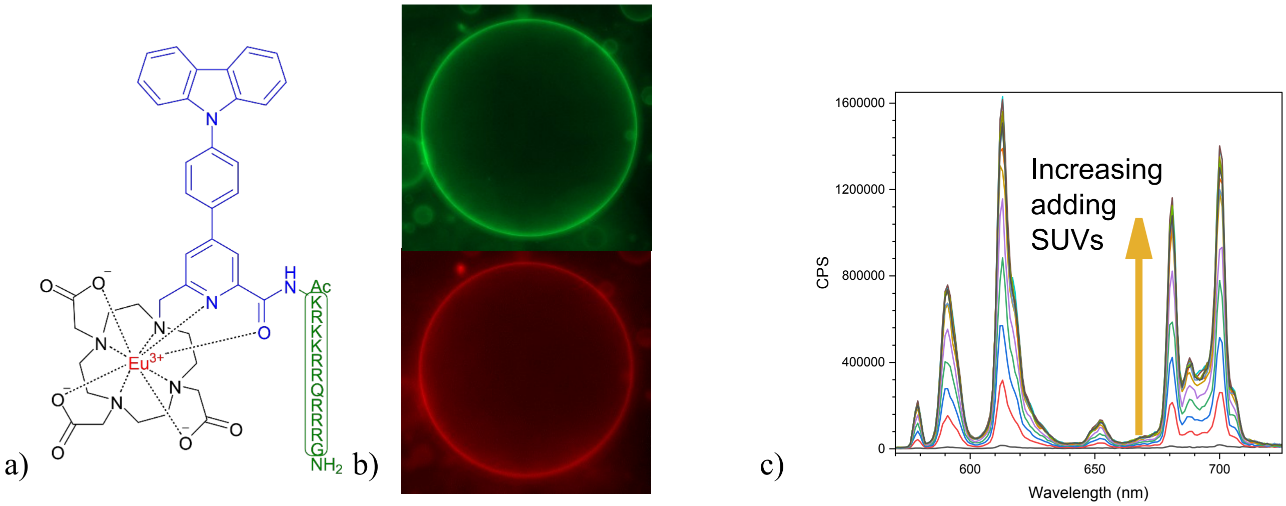
ABSTRACT

Lanthanide complexes exhibit unique luminescent properties: long emission lifetimes (ms), intense and narrow characteristic emission bands (which enable the study of the environment), as well as tunable absorption characteristics.[1]

The Eu^{3+} mTAT[Eu·L-Ar-Cz] complex (see Figure 1a), which contains an (N-carbazolyl)-aryl-picolinamide antenna, is conjugated to a polycationic TAT peptide capable of penetrating cells. The quantum yield of Eu^{3+} emission in this complex is highly dependent on solvent polarity, ranging from 0.13 in DMF to 0.0017 in PBS.[2] Nevertheless, this complex exhibited intense emission when incubated with cells, particularly in lipid-rich regions.

The interactions between the complex and the membrane were examined using epifluorescence microscopy and Giant Unilamellar Vesicles (GUVs) prepared from a DOPC/DOPG (90/10) mixture, revealing that the complex was colocalized at the membrane surface. A spectroscopic and small-angle neutron scattering study, carried out using liposomes of the same composition, showed that the complex is distributed across the surface of the membrane and that the antenna penetrates the membrane, thereby altering its local polarity and emission characteristics.



Fig– *a) Structure of mTAT[Eu·L-Ar-Cz] b) GUV (DOPC/DOPG 90:10): in green, NBD-lipid and in red, the complex c) Fluorescence emission of mTAT[Eu·L-Ar-Cz] in the presence of SUVs.*

REFERENCES

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